

Setting up your Workspace

Configuring the Brains and Brawn of your Development Environment:
Hugging Face and Google Colab

Hugging Face: The Github of AI

What is Hugging Face?

Hugging Face serves as the crossroads for accessing pre-trained models, datasets, and research demos.

> **Two-Way Repositories:** Just like Github, the Hugging Face Hub is a two-way street. You don't just download resources; you can **clone, commit, and push** your own custom models, fine-tuned weights, and datasets **directly** to the cloud.

> **Version Control:** Track model updates over time, manage branches, and collaborate with the open-source community using **pull requests** and **discussion threads**.

> **Authentication:** Securely manage your private repositories or request access to gated, industrial research models (like Meta Llama) using secure runtime tokens.



Setting Up Your Account



Join Hugging Face

Join the community of machine learners!

Email Address

Hint: Use your organization email to easily find and join your company/team org.

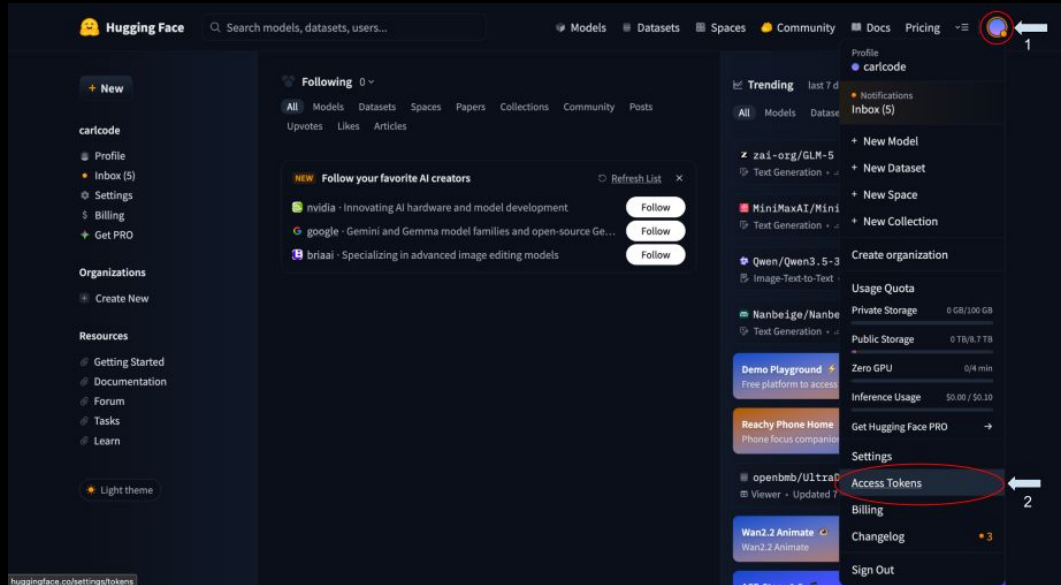
Password

Next

Already have an account? [Log in](#)

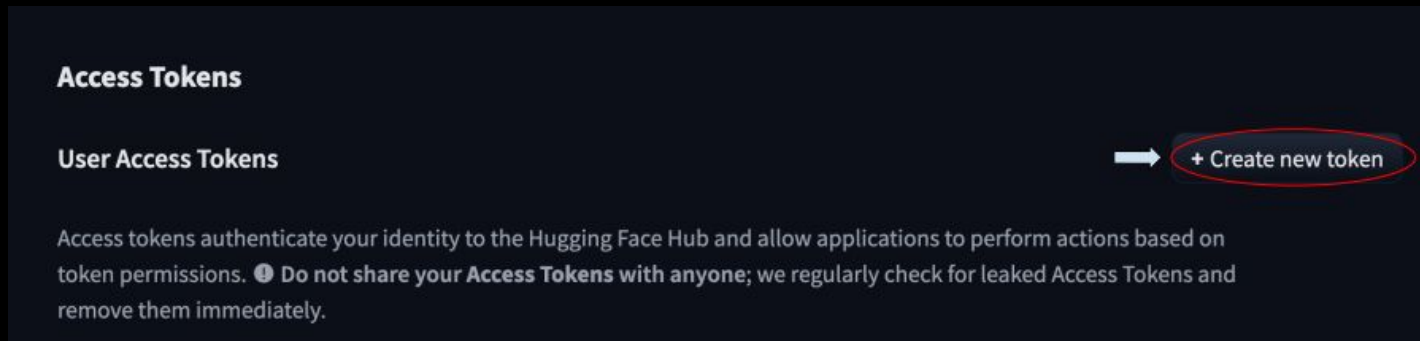
1. Visit the register page at:
<https://huggingface.co/join>
2. Enter your email and choose a password.
Note: If you are part of an organization we recommend using your organization email.
3. Confirm your email address via the link sent to your inbox. You will need to do this to enable **git** operations later.

Obtaining your User Access Token



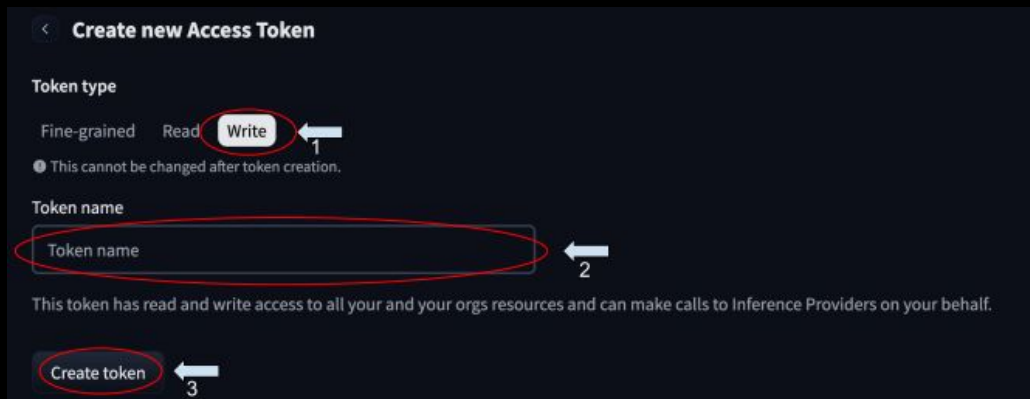
Log in to your Hugging Face account and click on your **Profile Icon** in the top-right corner of the page and select **Access Tokens** from the dropdown menu.

Obtaining your User Access Token Cont.



Click the [+ Create new token] button

Obtaining your User Access Token Cont.



The screenshot shows the 'Create new Access Token' page on Hugging Face. It features three main sections: 'Token type', 'Token name', and a 'Create token' button. Red circles and arrows with numbers 1, 2, and 3 highlight the 'Write' permission, the 'Token name' input field, and the 'Create token' button, respectively.

Create new Access Token

Token type

Fine-grained Read **Write** ← 1

⚙ This cannot be changed after token creation.

Token name

Token name ← 2

This token has read and write access to all your and your orgs resources and can make calls to Inference Providers on your behalf.

Create token ← 3

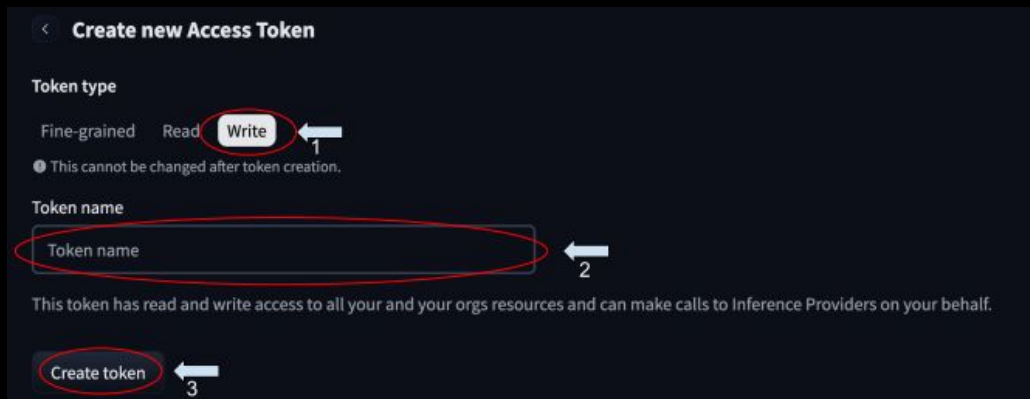
Define Permissions

> **Read:** Best for general research and downloading public or gated models.

> **Write:** Necessary for saving your fine-tuned model weights to the cloud. (We will be using this for this course)

> **Fine-Grained Tokens:** Allow you to specify exactly which repositories a notebook can access. This is recommended for professional researchers and teams to prevent accidental data loss or unauthorized access.

Obtaining your User Access Token Cont.



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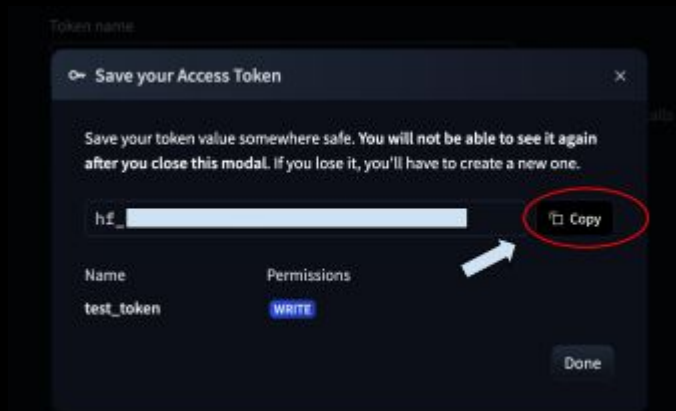
This token has read and write access to all your and your orgs resources and can make calls to Inference Providers on your behalf.

Create token ← 3

Give your token a clear, recognizable name.

Then, click on `[Create Token]`

Obtaining your User Access Token Cont.



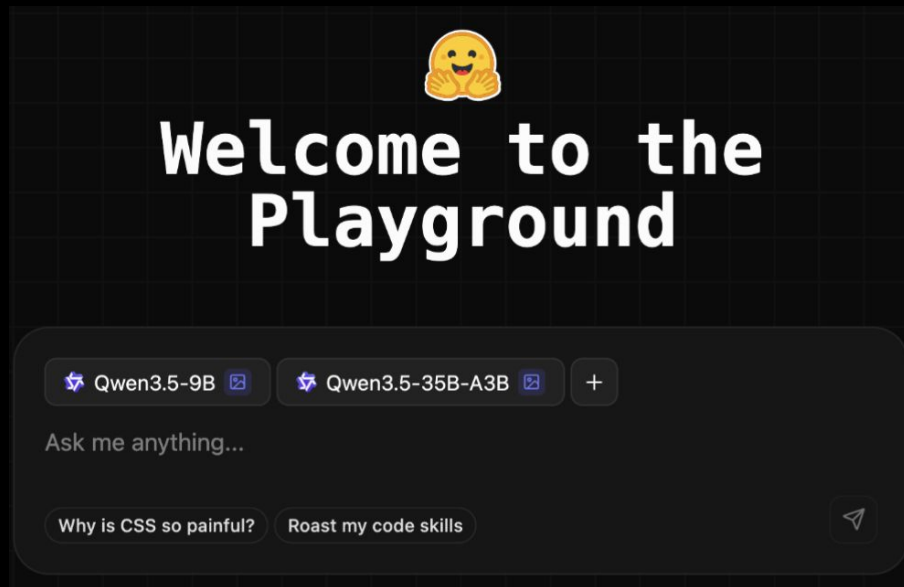
Copy the token value and place it somewhere `secure`.

Note: Never `hard-code` your token directly into a code cell. Use environment variables to prevent `unauthorized access`.

Using the Hugging Face Playground

What is it?

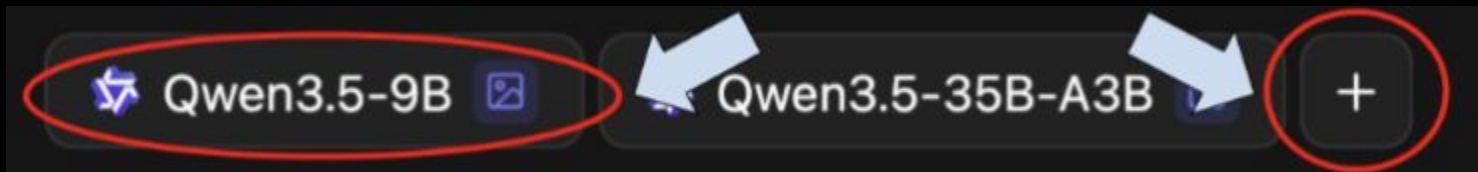
The Hugging Face Playground is an interactive LLM interface that allows you to test and experiment with different models directly in your browser, without writing any code!



To use the playground, navigate to:

<https://huggingface.co/playground>

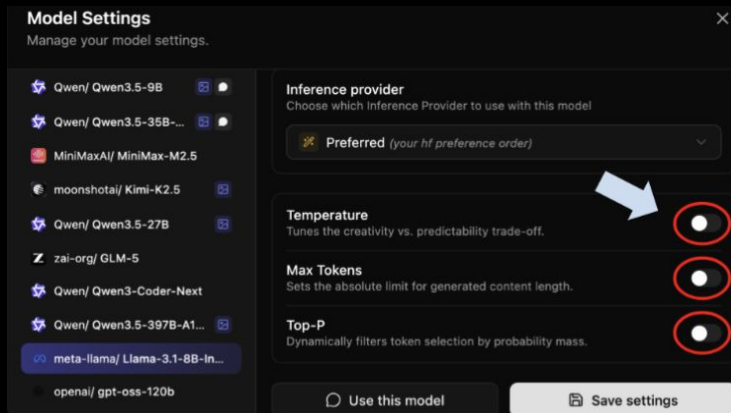
Using the Hugging Face Playground



Select one or more pre-trained models to interact with.

Note: If you select more than one model, the Playground will automatically set up a **side-by-side comparison** interface for you to test the **same prompt across different architectures**.

Using the Hugging Face Playground

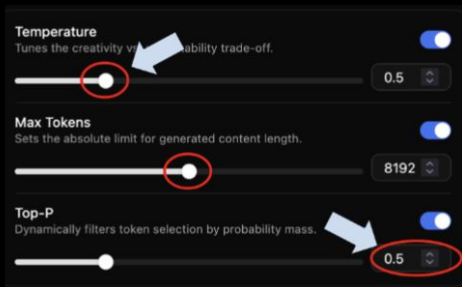


If you want to adjust the model's settings:

> Click on the model name above the text box to open the settings window.

> Here you can adjust parameters like temperature or top-p to see how they affect the model's output.

> Press Save to apply the changes.



Google Colab: On-Demand Cloud Compute

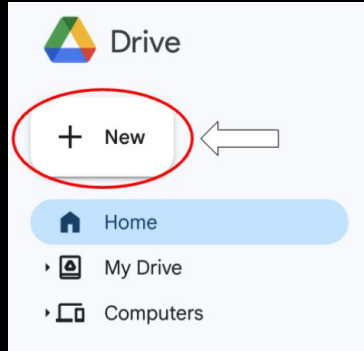
What is Google Colab?

Google Colab provides a browser-based Jupyter Notebook environment that requires zero local configuration.

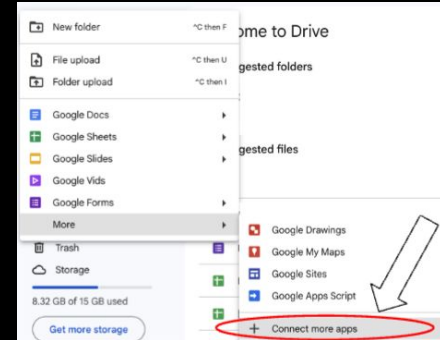
Its most important feature (for Generative AI) is access to cloud-allocated Hardware Accelerators like GPUs. This hardware is essential to handle the massive parallel math required for deep learning and fine-tuning.



Installing Google Colab

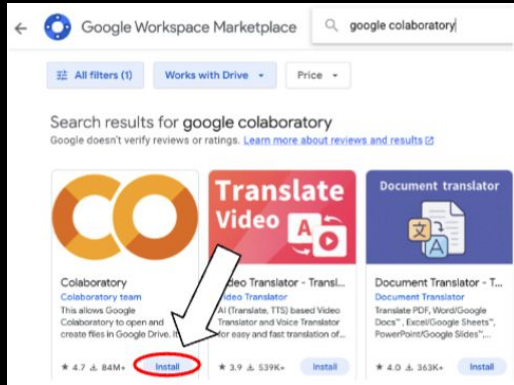


Log in to your **google account** and navigate to **google drive** and press **[+ New]**

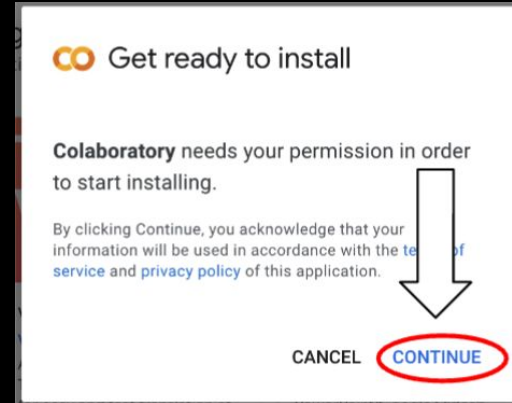


In the drop down menu, hover over **More** and press the **+ Connect more apps** option.

Installing Google Colab Cont.

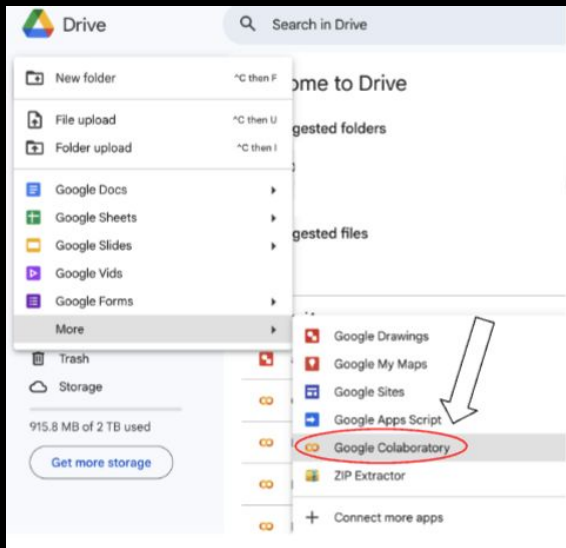


Search for **Colaboratory**



Install the application

Creating Your Colab Notebook



Navigate back to google drive, select the [+ New] button, hover over **More** and select **Google Colaboratory**.



To verify your connection to the Google runtime, into the first cell type:

```
print("Hello World")
```

And press Shift + Enter to run the cell. The output will appear below the code.

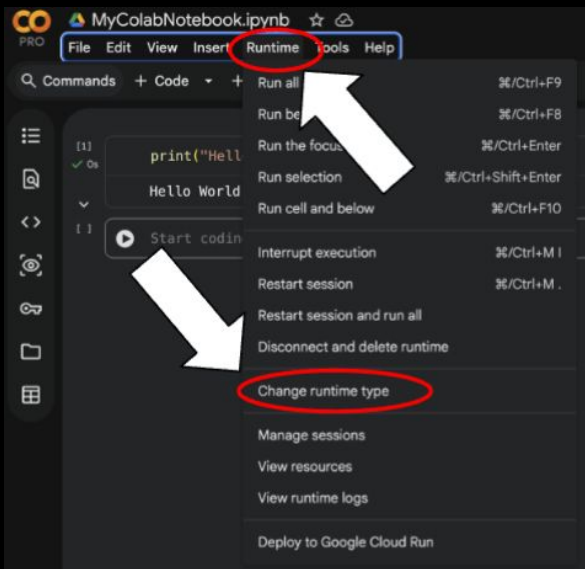
Configuring the Cloud GPU

Google Colab allows you to connect your notebook to a high-performance cloud GPU.

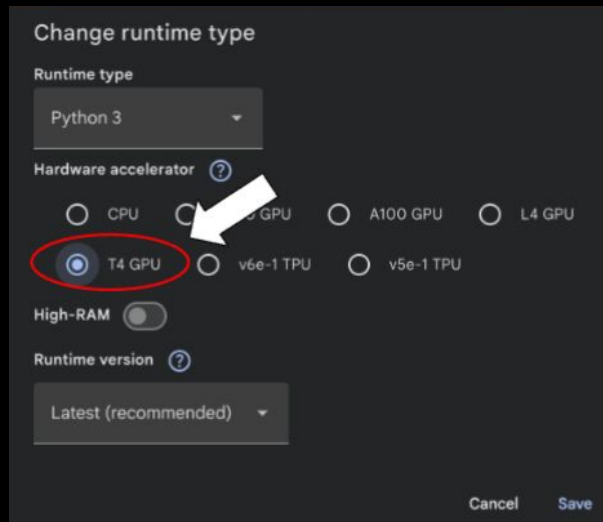
On the free tier, users are given a NVIDIA T4 GPU. This is what we will be using during the course.



Configuring the Cloud GPU Cont.

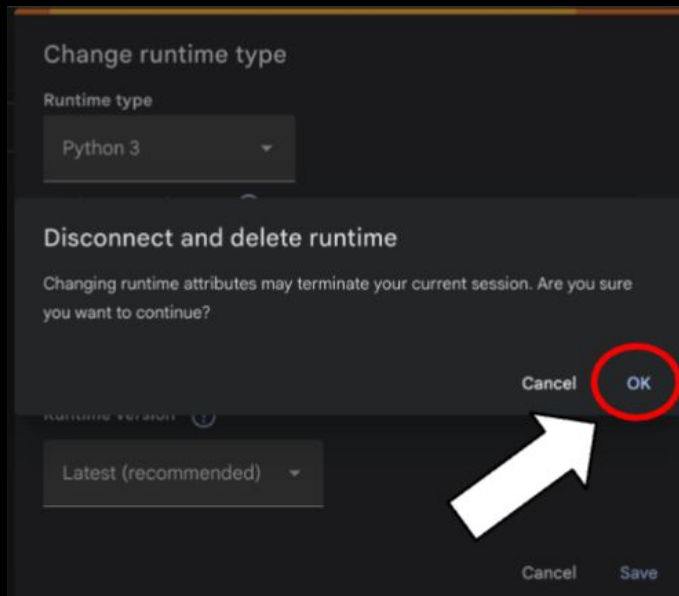


In your Colab notebook click on Runtime → Change runtime type



In the hardware accelerator section select T4 GPU.

Configuring the Cloud GPU Cont.



If prompted with a window stating "Disconnect and delete current runtime" click OK.

This clears the current session to make room for the GPU.

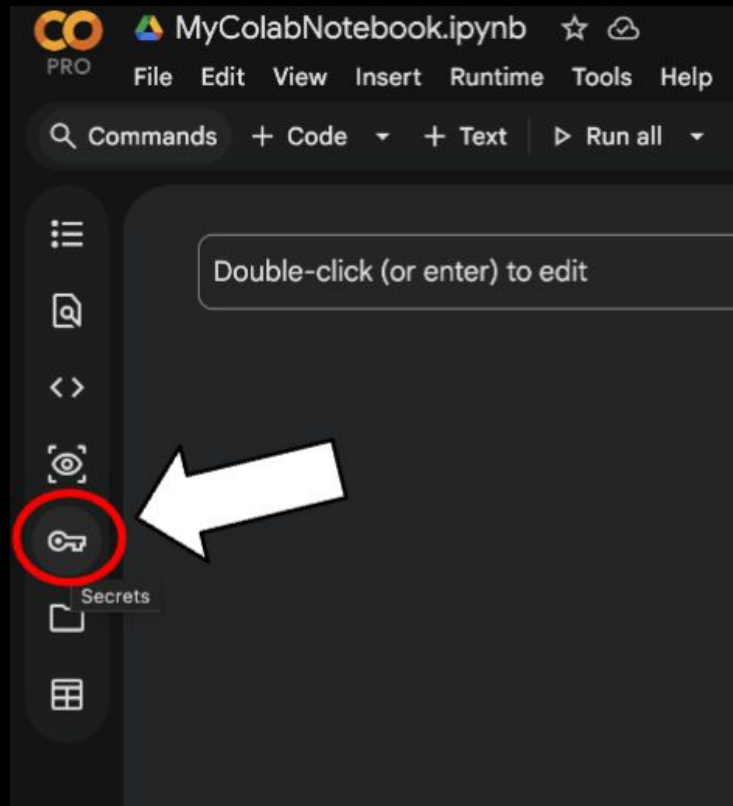
Note: On the free tier, GPU access is subject to availability and usage limits. If the T4 option is unavailable, you may need to wait for a new session or check your current usage quota.

Adding your Access Token to Secrets

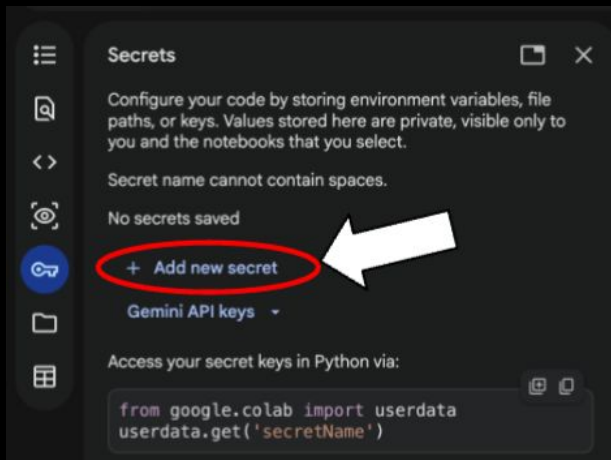
In order to maintain a secure and professional workflow, you should avoid pasting sensitive credentials directly into code cells.

Google Colab's **Secrets** manager allows you to store your Hugging Face Access Token as an environment variable that is isolated from the notebook's visible content.

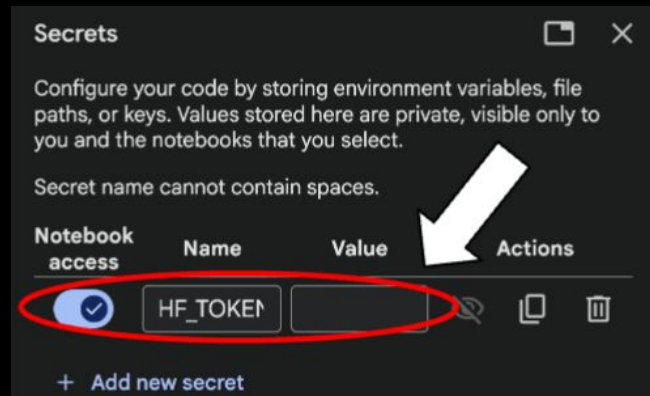
To add your HF Access Token to Colab Secrets, first navigate to the left-hand sidebar and select the **Key** icon



Adding your Access Token to Secrets Cont.



Then click on + Add new secret



Set the name of the secret to exactly:

HF_TOKEN

Then, set the value of the secret to your Hugging Face Access Token.

Make sure Notebook access is toggled "On"

Adding your Access Token to Secrets Cont.

Now that your secret is stored, run the following two lines:

```
from huggingface_hub import notebook_login
notebook_login()
```

If the cell runs without any errors or widgets popping up, you have successfully added your HF Access Token to your secrets!



Copy a token from [your Hugging Face tokens page](#) and paste it below.

Immediately click login after copying your token or it might be stored in plain text in this notebook file.

Token:

☒ Add token as git credential?

Login

Note: If this widget pops up either:

> Your Secret is not named exactly HF_TOKEN (case-sensitive)

> The "Notebook access" toggle is "Off"

Either way, you could also just paste your HF Token in the widget and your notebook will still have access to all the permissions your token holds.

Happy Coding!

```
=====
( ) ( ) ( ) ( ) ( ) ( ) ( ) ( )
( ) ( ) ( ) ( ) ( ) ( ) ( ) ( )
( ) ( ) ( ) ( ) ( ) ( ) ( ) ( )
      WORKSPACE INITIALIZED
=====

🔍 RUNTIME VERIFICATION MATRIX:
├── [SYSTEM]: Python 3.12.13
├── [COMPUTE]: ✓ CUDA Active (Tesla T4)
├── [STORAGE]: ✓ Google Drive Mounted
└── [NETWORK]: ✓ Hugging Face Hub Handshake Success (HF_TOKEN Linked)

System warm. Ready for Tokenization and Training.
```